Process Characterization Report

A-Mab Drug Substance — Production Bioreactor through Ultrafiltration/Diafiltration

Process Development / Manufacturing Science & Technology (MSAT)

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Executive summary

This report documents the process characterization (Stage 1, Process Design) of the A-Mab drug-substance manufacturing process, a humanized IgG1 monoclonal antibody whose principal mechanism of action is antibody-dependent cell-mediated cytotoxicity (ADCC). The work follows the lifecycle approach to process validation of the FDA 2011 guidance, PDA Technical Report 60 (Parenteral Drug Association 2013) and the ISPE Good Practice Guide (International Society for Pharmaceutical Engineering 2023), and applies the Quality-by-Design principles of ICH Q8, Q9 and Q11 (International Council for Harmonisation 2009, 2023, 2012). The methodology and quality-attribute framework are those of the A-Mab case study (CMC Biotech Working Group 2009).

Eight unit operations were modelled and characterized — production bioreactor, harvest, Protein A capture, low-pH viral inactivation, cation-exchange (CEX) and anion-exchange (AEX) chromatography, small-virus retentive filtration, and ultrafiltration/diafiltration (UF/DF). Scale-down models were used to execute screening and response-surface designs of experiments (DoE) that relate process parameters to critical quality attributes (CQAs) and to define proven acceptable ranges (PARs), design spaces and the control strategy.

Key outcomes

- **37 process parameters** were classified. **21** are critical (CPP or well-controlled CPP, WC-CPP); the remainder are key (KPP) or general (GPP) process parameters.
- All drug-substance CQAs meet their acceptance criteria at set-point, and a Monte-Carlo simulation of **2,000 commercial-scale batches** gives a minimum process-capability index of **Cpk = 1.51** across all CQAs.
- Cumulative viral clearance is **18.87 log (XMuLV, requirement > 16.7)** and **10.03 log (MVM, requirement > 8.6)**.
- Overall product step-yield is **83.2%**, giving **54.6 kg** of drug substance per 15,000 L batch at 75.0 g/L.
- The accompanying post-characterization FMEA shows the median risk priority number falling by ~85% once the characterized ranges and control strategy are in place — the quantitative demonstration of enhanced process understanding.

The process is demonstrated to be well understood and robust, and is considered ready to proceed to Stage 2 (Process Performance Qualification).

1 Purpose and scope

Purpose. This report is the Process Design Report for A-Mab drug substance as defined in PDA TR 60 §3.11 (Parenteral Drug Association 2013). It consolidates the critical quality attributes and their supporting risk assessments, the process description and parameter ranges, the classification of each parameter for its risk of impact to CQAs and process performance, the design spaces established by multivariate experimentation, and the justification and data supporting every parameter range. It is the repository that links the risk assessments, characterization studies and control strategy, and it provides the justification for the Stage 2 (PPQ) ranges and acceptance criteria.

Scope. The drug-substance process, Steps 3–10 (production bioreactor through UF/DF), operated at the 15,000-L commercial scale. Seed-train expansion (Steps 1–2) is addressed only where it was shown capable of affecting the production bioreactor (Section 4). UF/DF is included for mass balance; its formulation characterization is reported separately with the drug product. Analytical method validation, facility/equipment qualification and drug-product processing are out of scope.

Raw materials and critical material attributes (CMAs). Consistent with the A-Mab platform approach, raw-material and cell-culture-medium attributes are controlled to established platform specifications and are treated as identity/quality-controlled inputs rather than characterized variables in this study; where a material attribute was a potential CQA driver (e.g. medium concentration) it was included in the bioreactor characterization (Section 6.1). Formal CMA characterization and supplier controls are managed under the raw-material control programme and are out of scope of this report.

Basis. The process model, quality-attribute framework and risk methodology follow the A-Mab case study (CMC Biotech Working Group 2009); the report structure and statistical approach follow PDA TR 60 (Parenteral Drug Association 2013) and the ISPE Good Practice Guide (International Society for Pharmaceutical Engineering 2023); the QbD framework follows ICH Q8/Q9/Q11 (International Council for Harmonisation 2009, 2023, 2012).

2 Product and process background

2.1 Quality target product profile (QTPP)

Table 2.1: Quality target product profile for A-Mab.

Attribute	Target
Dosage form	Sterile liquid for intravenous infusion
Drug substance	Humanized IgG1 mAb, 75.0 g/L, iso-osmolar formulation
Mechanism of action	Antibody-dependent cell-mediated cytotoxicity (ADCC); CDC contributory
Manufacturing scale	15,000 L fed-batch production bioreactor
Purity / safety	Aggregates, HCP, DNA, leached Protein A within acceptance; validated viral clearance
Potency	ADCC 70–130% of reference; governed by Fc glycosylation

2.2 Process description and flow

The A-Mab drug substance is produced by a platform monoclonal-antibody process (Figure 2.1). Product quality is established in the production bioreactor (glycosylation, charge variants and the initial size-variant and impurity load) and is subsequently purified: Protein A captures the antibody and clears the bulk of host-cell protein (HCP) and DNA; low-pH viral inactivation provides enveloped-virus safety; CEX is the principal aggregate- and HCP-polishing step; AEX in flow-through mode removes residual HCP, DNA and leached Protein A and provides an orthogonal viral-clearance claim; small-virus retentive filtration provides a size-based, modular viral-clearance claim; and UF/DF concentrates and formulates the drug substance.

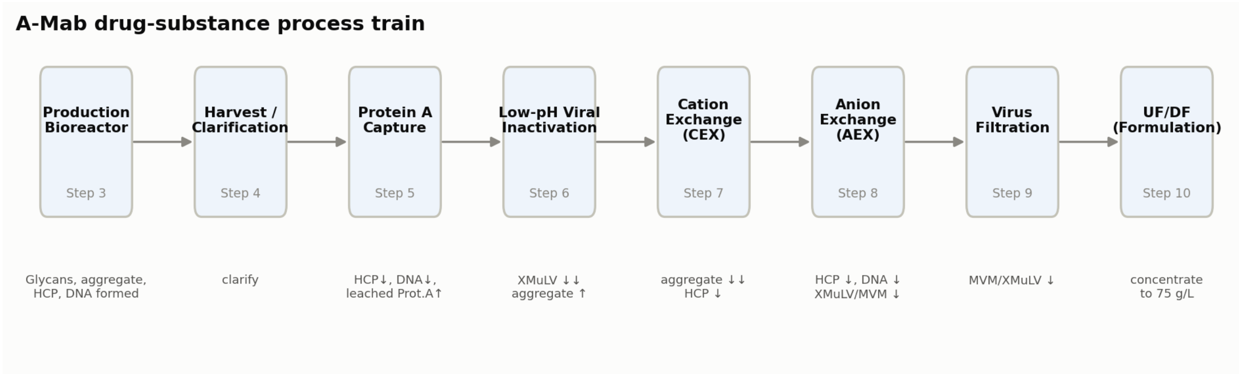

Figure 2.1: A-Mab drug-substance process flow (Steps 3–10) with the principal quality impact of each unit operation.

2.3 Process description, in-process controls and outputs

Each unit operation is summarized in Table 2.2 (mode/scale, estimated step yield and the principal hold time/condition). The inputs to each step are the process parameters and their ranges (Section 6); the step *outputs* — in-process tests, limits and in-process specifications — are given in Table 2.3. Final drug-substance acceptance criteria and the statistically-justified PPQ sampling plans are established in Stage 2 (Section 8).

Table 2.2: Process description and estimated per-step performance.

Step	Unit operation	Mode / scale	Est. step yield	Principal hold / condition
3	Production Bioreactor	Fed-batch cell culture, 15,000 L	100%	15–19 day culture; harvest on viability/titer/quality
4	Harvest / Clarification	Centrifugation + depth/sterile filtration	98%	24 h post-harvest hold, 2–8 °C
5	Protein A Chromatography	Protein A affinity, bind–elute, multi-cycle	97%	Low-pH eluate held for viral inactivation
6	Low-pH Viral Inactivation	Low-pH batch incubation	99%	pH 3.5, 60–120 min, 15–25 °C
7	Cation Exchange Chromatography	Cation exchange, bind–elute	95%	Eluate pool hold 24 h, 2–8 °C
8	Anion Exchange Chromatography	Anion exchange, flow-through	97%	Single-cycle flow-through
9	Small Virus Retentive Filtration	Small-virus retentive filtration	98%	Volumetric load 105 L/m²
10	Ultrafiltration / Diafiltration	Ultrafiltration/diafiltration (TFF)	97%	Formulate to 75 g/L; DS hold 2–8 °C

Table 2.3: In-process controls, tests and outputs by unit operation (representative alert/action limits; final specifications set in Stage 2).

Unit operation	In-process test / output	Alert / action limit or in-process specification
Production Bioreactor	Culture viability & titer at harvest	Harvest criteria on viability, titer and quality
Harvest	Post-clarification turbidity; bioburden	Turbidity trended; bioburden action limit per programme
Protein A	Step yield (A280); pool HCP (trend)	Yield 75%; HCP trended
Low-pH Viral Inactivation	Inactivation pH; hold time	pH 3.4–3.6; hold 60 min (action)
Low-pH Viral Inactivation	Pool aggregate (SEC)	3.1% (action)
CEX	Step yield; pool aggregate (SEC)	Yield target; aggregate release-linked limit
AEX	Eluate endotoxin	Alert 5 EU/mL; action 10 EU/mL
AEX	Eluate bioburden	Alert 10 CFU/10 mL; action 50 CFU/10 mL
AEX	Pool HCP; residual DNA	HCP trended to release limit; DNA < LOQ
Virus Filtration	Pre/post-use filter integrity test; volumetric load	Integrity test: pass; load 105 L/m^2
UF/DF	Drug-substance concentration; endotoxin/bioburden	70–80 g/L; per DS specification

3 Critical quality attributes and criticality

CQAs were identified and their criticality assessed on a continuum (ICH Q9) using the A-Mab Tool #1 risk score = Impact × Uncertainty, where impact is scored on efficacy/biological activity, PK/PD, immunogenicity and safety, and uncertainty reflects the strength of the supporting knowledge (CMC Biotech Working Group 2009). Attributes rated High (H) or Very High (VH) are treated as *Critical*. Table 3.1 summarizes the CQAs, their acceptance criteria and criticality; Figure 2.1 indicates where each is set or controlled.

Table 3.1: A-Mab critical quality attributes and criticality (Tool #1 = Impact × Uncertainty; H/VH = Critical).

CQA	Category	Acceptance	Criticality	Tool#1 score	Set/controlled by
Afucosylation	glycosylation	2–13 % of glycans	H	60	bioreactor
Galactosylation (total %Gal)	glycosylation	10–40 % (G1+G2 equiv.)	H	48	bioreactor
High Mannose	glycosylation	3–10 % of glycans	VH	80	bioreactor
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60	bioreactor
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4	bioreactor
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36	bioreactor
Residual DNA	process impurity	0–0.001 ng/dose	VL	6	bioreactor
Leached Protein A	process impurity	0–5 ppm	L-M	16	protein_a
Viral clearance — XMuLV (enveloped)	viral safety	16.7–30 log10 (cumulative)	VH	20	viral_inactivation
Viral clearance — MVM (parvovirus)	viral safety	8.6–20 log10 (cumulative)	VH	20	aex

The glycosylation attributes (afucosylation, high mannose, non-glycosylated heavy chain, galactosylation) are the most critical because they govern ADCC and CDC potency; they are determined in cell culture and are not modified by downstream processing. Aggregates and HCP are managed by the purification train. Viral safety is assured by process design (three orthogonal clearance steps) because it cannot be measured in the final product.

4 Risk-assessment strategy

Process characterization was directed by a lifecycle of iterative quality risk assessments (ICH Q9), consistent with A-Mab Risk Assessments #1–#4 (CMC Biotech Working Group 2009) and the Stage-1 risk flow of PDA TR 60 (Parenteral Drug Association 2013):

1. **Initial risk assessment (RA#1–#2)** — prior platform knowledge and early development experience prioritized the process steps and parameters carrying the greatest potential impact to CQAs, and set the characterization scope (which parameters, which ranges, which interactions).
2. **Process characterization** — screening DoE (two-level factorial / resolution-V fractional factorial) identified the significant parameters; response-surface designs (face-centred central-composite) quantified main effects, interactions and curvature and defined the design space.
3. **Post-characterization risk assessment (RA#3–#4)** — the study results re-scored the process risk and fixed the parameter classification and control strategy. This is the subject of the accompanying post-PC FMEA (RPN = Severity × Occurrence × Detection). **CPP status is set by the pre-control risk of impacting a CQA** — a parameter that affects a CQA with Severity 8 **or** an initial (pre-mitigation) RPN > 72 is a CPP. The control strategy then determines whether that CPP is a *well-controlled CPP* (WC-CPP, low risk of leaving the design space) or remains a CPP requiring the most stringent control. The *residual* RPN reflects the post-control Occurrence and Detection; a high-severity attribute retains an elevated residual RPN even when well controlled — which is precisely why it is designated (WC-)CPP and enhanced-controlled.

Parameter designations follow the A-Mab continuum: **CPP** (variability impacts a CQA; high risk of leaving the design space), **WC-CPP** (impacts a CQA but well controlled, low risk), **KPP** (impacts process performance/consistency, not product quality), and **GPP** (general parameter, no meaningful impact over a wide range).

Prior knowledge and manufacturing history. The characterization ranges are supported not only by the scale-down studies reported here but by prior platform knowledge from related antibodies (X-Mab, Y-Mab, Z-Mab) and by A-Mab clinical and engineering manufacturing experience — the basis for the CQA acceptance ranges (Section 3), the modular viral-clearance claims (Sections 6.4, 6.6, 6.8) and the set-point selection. For a regulatory filing this section would summarize the clinical, engineering and process-evaluation batch history (batch records, campaign summaries and comparability data) that corroborates each proven acceptable range; that history is referenced here and maintained in the development report set.

5 Scale-down model qualification

All characterization studies were performed on qualified scale-down models (SDMs): a 2 L bioreactor for the production culture and bench-scale columns/devices for the downstream steps, each operated at the commercial bed height, linear velocity and load/wash/elution column-volumes (or membrane device equivalents). SDM qualification compared operational parameters and, critically, the input and output quality attributes (titer, viability, HCP, DNA, aggregate, charge and glycan profiles, and log-reduction values) against at-scale reference data, confirming the models are representative and suitable to support design-space and PAR claims (Parenteral Drug Association 2013). Viral-clearance claims additionally leverage the A-Mab modular approach across related antibodies.

6 Process characterization studies

Each unit operation was characterized against the CQAs and performance attributes it affects. For each step the following subsections give the study design, the parameters and ranges studied, the significant effects, the resulting design space, and the parameter classification.

6.1 Production bioreactor (Step 3)

The production bioreactor establishes the glycosylation, charge-variant and initial size-variant/impurity profile of A-Mab and is the sole upstream step used to define the upstream design space. A five-factor resolution-V screening design followed by a face-centred central-composite response-surface design characterized culture pH, temperature, dissolved CO , osmolality and culture duration (dissolved oxygen, inoculation density, medium concentration and feed volume were assessed for process performance).

Table 6.1: Production-bioreactor process parameters, ranges and classification.

Parameter	Unit	Set-point	NOR	PAR	Class	Study
Culture pH	pH	6.85	6.75–6.95	6.6–7.1	WC-CPP	multivariate
Culture temperature	°C	35	34.5–35.5	34–36	WC-CPP	multivariate
Dissolved CO2 (pCO2)	mmHg	100	70–130	40–160	WC-CPP	multivariate
Osmolality	mOsm	400	375–425	360–440	WC-CPP	multivariate
Culture duration	days	17	16–18	15–19	WC-CPP	multivariate
Dissolved oxygen	% sat	50	40–60	30–70	KPP	multivariate
Initial viable cell conc.	e6 cells/mL	1	0.8–1.2	0.5–1.5	KPP	univariate
Basal medium concentration	X	1.2	1–1.4	0.8–1.6	GPP	multivariate
Nutrient feed-1 volume	% WV	12	10–14	9–15	KPP	multivariate

The nominal fed-batch profile (Figure 6.1) reaches a peak viable cell density of 13.48 ×10 cells/mL and a harvest titer of 4.37 g/L at 67.1% final viability. Screening identified culture pH, temperature, CO , osmolality and duration as significant for the glycosylation and charge-variant CQAs (Figure 6.2, Table 6.2); the response-surface model reproduces the multivariate, interacting behaviour of the A-Mab design space (CMC Biotech Working Group 2009).

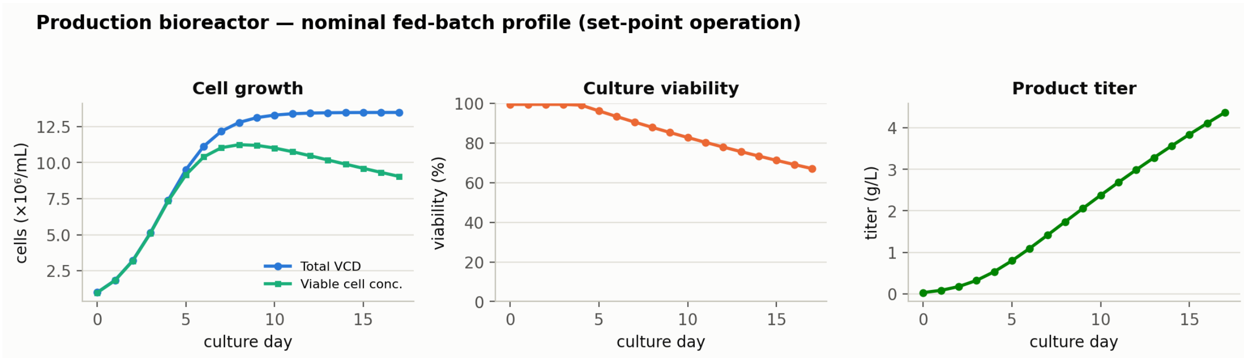

Figure 6.1: Nominal production-bioreactor time course: viable cell density, viability and titer.

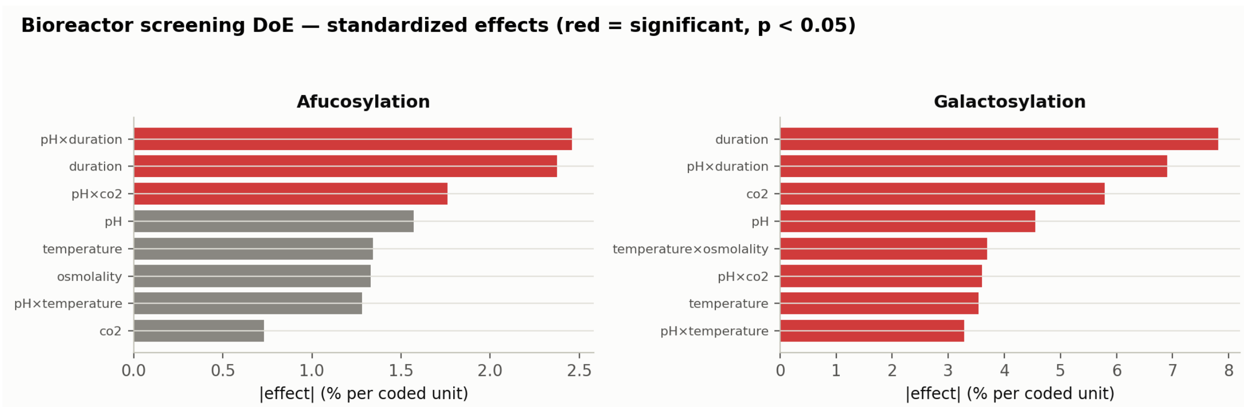

Figure 6.2: Standardized effects (screening DoE) for afucosylation and galactosylation; red bars are significant ($p < 0.05$).

Table 6.2: Largest standardized effects on afucosylation (bioreactor screening DoE).

Term	Effect	p-value	signif.
pH × duration	2.46	0.0191	yes
duration	-2.37	0.0209	yes
pH × co2	-1.76	0.0455	yes
pH	-1.57	0.06	
temperature	-1.34	0.0858	
osmolality	-1.33	0.0878	

Lower pH and lower temperature increase specific productivity, culture duration and titer, with a modest increase in afucosylation and a decrease in galactosylation; extended culture duration lowers afucosylation and galactosylation and raises the harvest HCP and DNA load as viability declines. Within the characterized ranges all cell-culture CQAs remain within acceptance simultaneously, defining a robust multivariate design space (Figure 6.3). Culture pH, temperature, CO , osmolality and duration are classified **WC-CPP** (significant CQA effects, but reliably controlled within the design space); dissolved oxygen, inoculation density and feed volume are **KPP**; medium concentration is **GPP**.

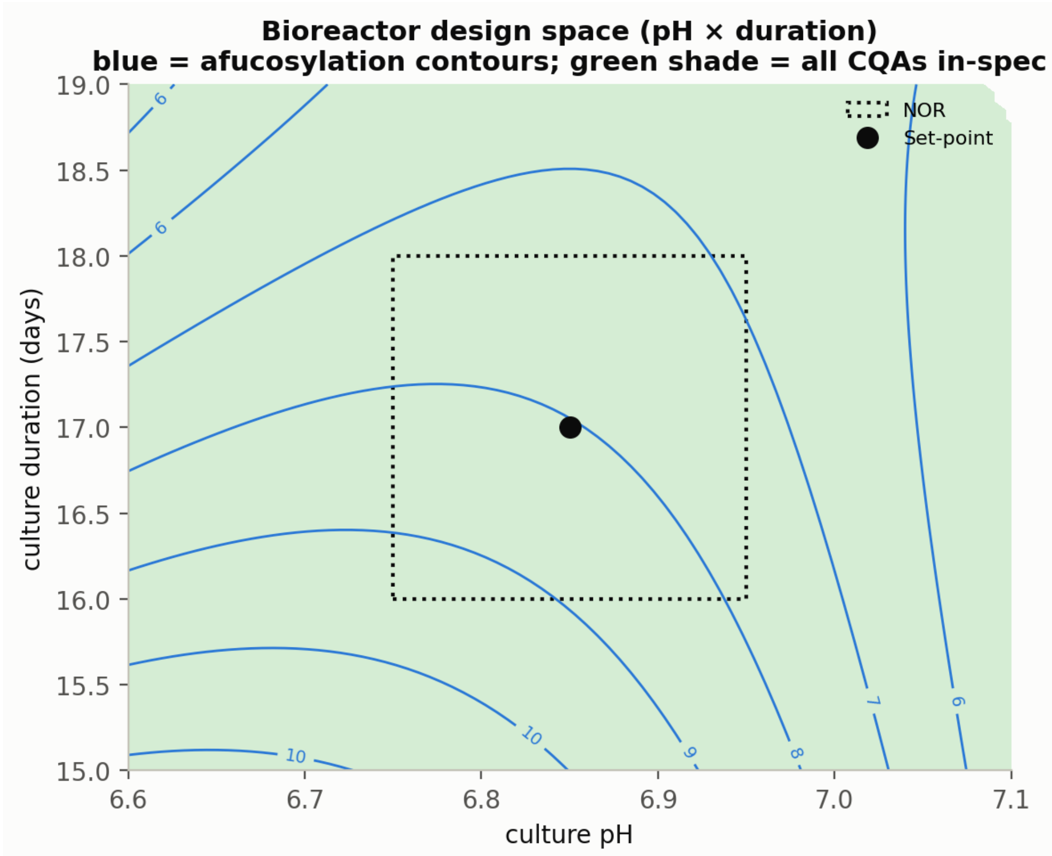

Figure 6.3: Production-bioreactor design space (pH × culture duration); the shaded region satisfies all cell-culture CQAs simultaneously. Blue contours are afucosylation (%).

6.2 Harvest and clarification (Step 4)

Harvest (centrifugation followed by depth and sterile filtration) clarifies the culture and defines the feed to Protein A. No product-quality impact was demonstrated; the step parameters (centrifugation g-force, depth-filter load, post-clarification turbidity) are KPP/GPP controlled for process performance and feed consistency (CMC Biotech Working Group 2009).

6.3 Protein A affinity chromatography (Step 5)

Protein A captures the antibody and clears the bulk of HCP and DNA. A six-factor resolution-IV screening design and a response-surface design characterized protein load, elution pH, load flow rate and end-of-pool collection.

Table 6.3: Protein A process parameters, ranges and classification.

Parameter	Unit	Set-point	NOR	PAR	Class	Study
Protein load	g/L resin	30	20–40	10–50	WC-CPP	multivariate
Elution buffer pH	pH	3.55	3.4–3.7	3.2–3.9	WC-CPP	multivariate
Load flow rate	cm/hr	200	150–250	100–300	KPP	multivariate
End of pool collect	CV	2.6	2.3–2.9	2–3.2	KPP	multivariate
Operating temperature	°C	22	18–26	15–30	GPP	multivariate
Bed height	cm	20	15–25	10–30	GPP	univariate

Pool HCP is the quality-determining response: it increases with higher protein load and lower elution pH (Figure 6.4), reproducing A-Mab Figure 4.2, while remaining well below the levels the CEX and AEX steps subsequently clear. Load flow rate and end-of-pool collection affect product yield (high load × high flow reduces recovery) but not quality. Protein load and elution pH are therefore **WC-CPP** (impact HCP, controlled within the design space); flow rate and end-collection are **KPP**; the remaining parameters are **GPP**.

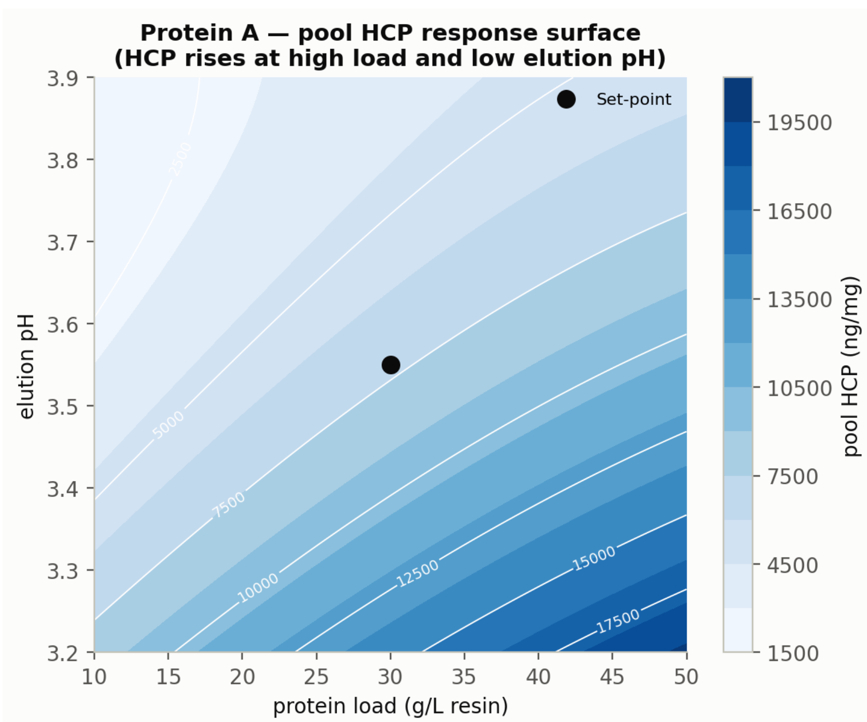

Figure 6.4: Protein A pool-HCP response surface (load × elution pH): HCP rises at high load and low elution pH.

6.4 Low-pH viral inactivation (Step 6)

Low-pH incubation inactivates enveloped virus (XMuLV). pH, hold time, temperature and protein concentration were characterized against XMuLV log-reduction and product quality (aggregation).

Table 6.4: Low-pH viral inactivation process parameters, ranges and classification.

Parameter	Unit	Set-point	NOR	PAR	Class	Study
Inactivation pH	pH	3.5	3.4–3.6	3.2–4	CPP	multivariate
Hold time	min	90	60–120	60–180	WC-CPP	multivariate
Temperature	°C	21	19–23	15–25	WC-CPP	multivariate
A-Mab concentration	g/L	20	5–35	5–35	GPP	univariate

pH is the defining parameter: above pH 4.0 inactivation of XMuLV within the hold is not assured, while below pH 3.2 the antibody may aggregate or precipitate — a narrow, high-consequence range that makes pH a **CPP**. Aggregate increases modestly with hold time and temperature (Figure 6.5, left) but remains below acceptance across the range, and is further polished by CEX; the XMuLV log-reduction surface (Figure 6.5, right) exceeds the required clearance throughout the acceptable pH × time region. Hold time and temperature are **WC-CPP**; protein concentration (35 g/L) is **GPP**.

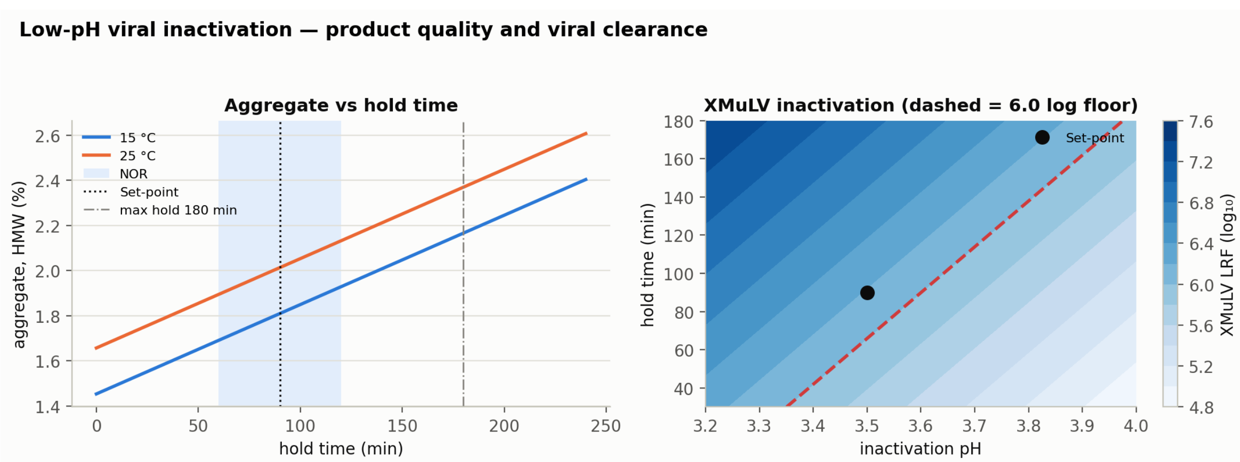

Figure 6.5: Low-pH viral inactivation: aggregate vs hold time at 15/25 °C (left) and the XMuLV log-reduction surface over pH × hold time (right).

6.5 Cation exchange chromatography — CEX (Step 7)

CEX operates in bind-and-elute mode and is the principal aggregate-polishing step; it also reduces HCP and clears DNA and leached Protein A. A fractional-factorial screen and a response-surface design characterized protein load, load/wash conductivity, elution pH and elution stop-collect.

Table 6.5: CEX process parameters, ranges and classification.

Parameter	Unit	Set-point	NOR	PAR	Class	Study
Protein load	g/L resin	25	18–32	10–40	WC-CPP	multivariate
Load/Wash conductivity	mS/cm	5	4–6	3–7	WC-CPP	multivariate
Elution buffer pH	pH	6	5.85–6.15	5.8–6.2	WC-CPP	multivariate
Elution stop collect	OD desc.	1	0.7–1.3	0.5–1.5	WC-CPP	multivariate
Elution flow rate	cm/hr	200	150–250	100–300	GPP	multivariate

Aggregate clearance (2–3×) weakens at higher protein load and wider elution stop-collect, and HCP clearance (50–100×) weakens at higher load and lower load/wash conductivity, with a significant load × conductivity interaction (Figure 6.6). Because there is no downstream aggregate-reduction step, a worst-case aggregate feed (linked to the low-pH hold) confirmed that CEX reduces aggregate below acceptance across all conditions. Protein load, load/wash conductivity, elution pH and stop-collect are **WC-CPP**; elution flow rate is **GPP**.

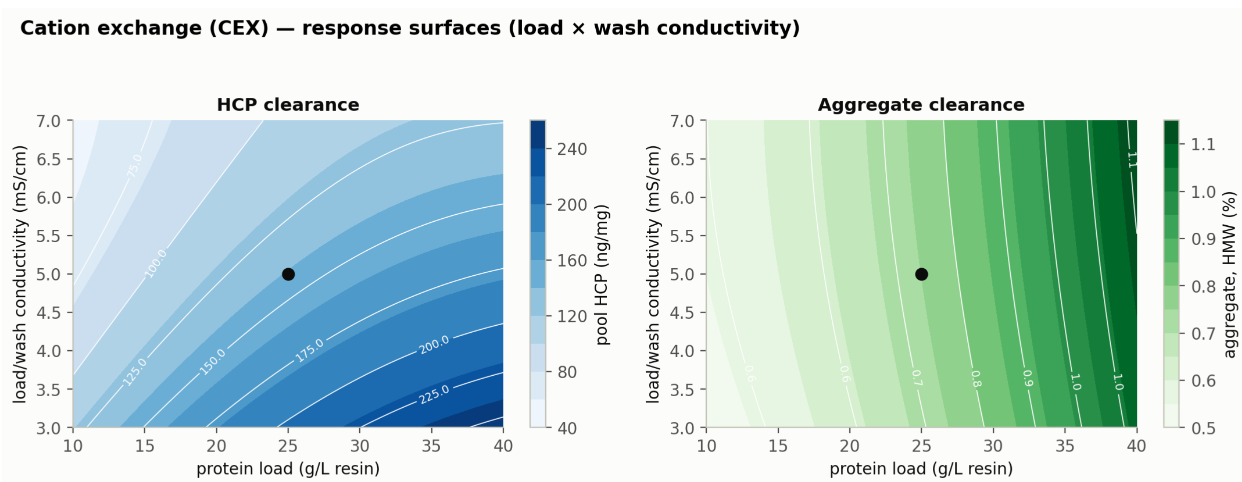

Figure 6.6: CEX response surfaces for pool HCP and aggregate over protein load × load/wash conductivity.

6.6 Anion exchange chromatography — AEX (Step 8)

AEX operates in flow-through mode: the antibody passes through while HCP, DNA, leached Protein A and endotoxin bind. AEX also provides an orthogonal viral-clearance claim (XMuLV and MVM). Load pH, Equilibration/Wash-1 conductivity, load conductivity and protein load were characterized.

Table 6.6: AEX process parameters, ranges and classification.

Parameter	Unit	Set-point	NOR	PAR	Class	Study
Load pH	pH	7.5	7.35–7.65	7.2–7.8	WC-CPP	multivariate
Equil/Wash-1 conductivity	mS/cm	2.6	2.1–3.1	1.6–3.6	WC-CPP	multivariate
Load conductivity	mS/cm	5.5	4–7	3–8	WC-CPP	multivariate
Protein load	g/L resin	200	120–280	50–300	WC-CPP	multivariate
Operating flow rate	cm/hr	300	200–400	150–450	WC-CPP	multivariate

HCP removal and viral clearance both decline as load pH decreases and conductivity increases (Figure 6.7); the quality design space (pH 7.2–7.8, Equil/Wash-1 conductivity 1.6–3.6 mS/cm) is narrower than the validated viral-clearance ranges, so operation is constrained by the quality requirement. Load pH, both conductivities, protein load and flow rate are **WC-CPP**.

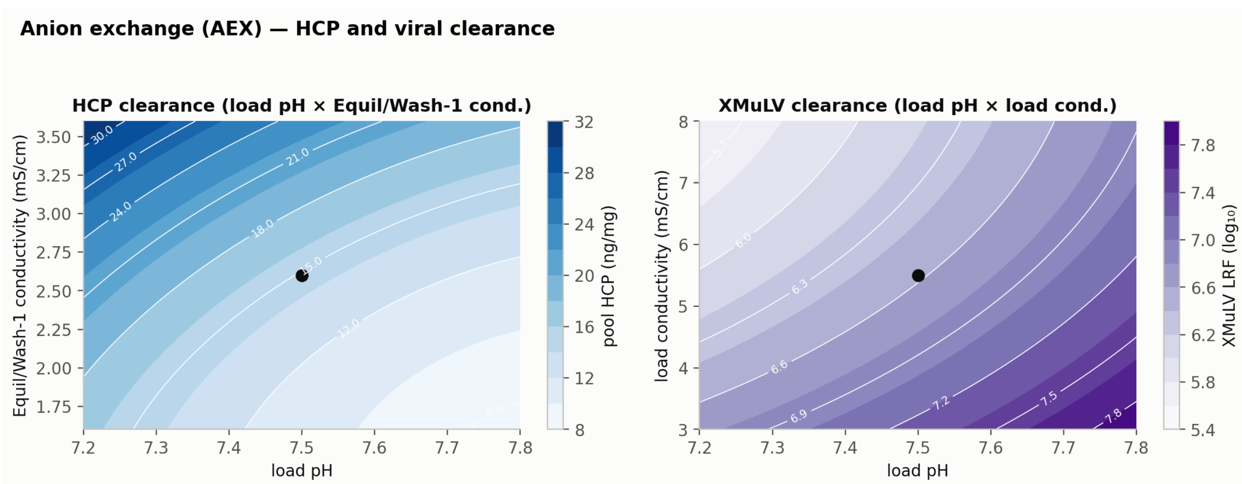

Figure 6.7: AEX response surfaces: HCP clearance and XMuLV log-reduction both fall as load pH decreases and Equil/Wash-1 conductivity increases.

6.7 Small-virus retentive filtration (Step 9)

Size-based filtration provides an orthogonal, modular clearance claim for small non-enveloped virus (MVM) and larger enveloped virus (XMuLV). MVM log-reduction declines with volumetric load; a load limit of 105 L/m² preserves LRV 4.62 (Figure 6.8). Filtration volume and pressure are **WC-CPP**; the pre/post-use integrity test is a procedural control.

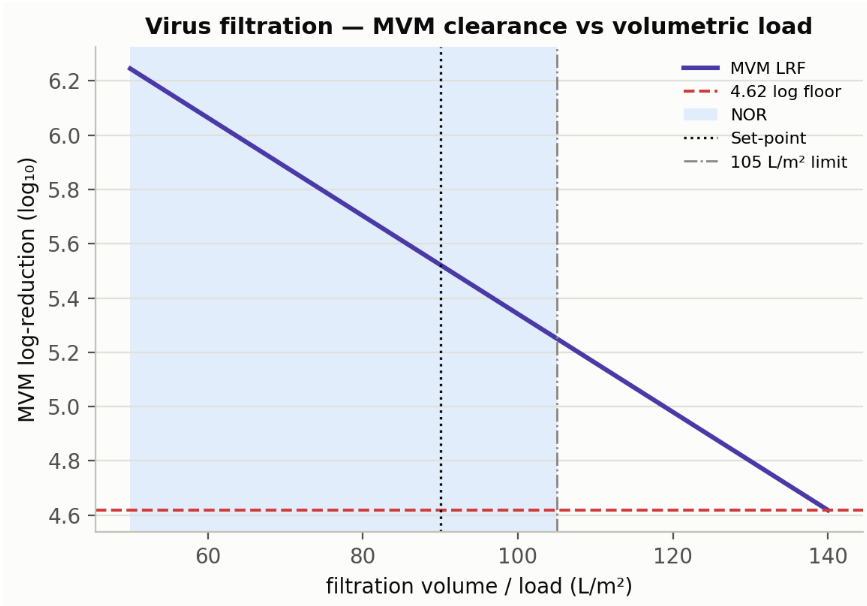

Figure 6.8: Small-virus filtration: MVM log-reduction vs volumetric load; the 105 L/m² limit preserves LRF   4.62.

6.8 Ultrafiltration / diafiltration (Step 10)

UF/DF concentrates and buffer-exchanges the antibody to the drug-substance target of 75.0 g/L. Consistent with the A-Mab case study, formulation characterization is reported with the drug product; the step is included here for mass balance only, and its parameters (diavolumes, transmembrane pressure, final concentration) are controlled as KPP.

7 Parameter classification summary

Across the drug-substance train, 37 process parameters were classified (Figure 7.1, Table 7.1). 21 are critical (CPP or WC-CPP), concentrated in the bioreactor (glycosylation and charge control), the chromatography steps (aggregate and HCP control) and the viral-safety steps.

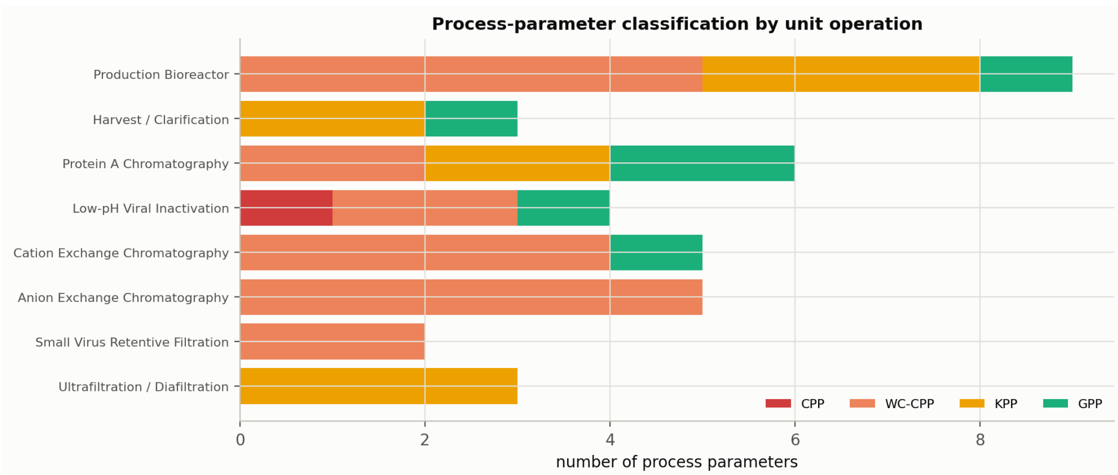

Figure 7.1: Post-characterization classification of process parameters by unit operation.

Table 7.1: Process-parameter classification summary.

Classification	Number of parameters
CPP	1
WC-CPP	20
KPP	10
GPP	6
Total	37

The complete post-characterization FMEA — one row per parameter with failure mode, severity/occurrence/detection scores, RPN before and after characterization, designation, control strategy and residual risk — is provided as the accompanying workbook, *A-Mab_Post-PC_Process_Risk_Assessment.xlsx*.

8 Process capability and Stage-2 readiness

A Monte-Carlo simulation of 2,000 commercial-scale batches, with every parameter varied within its normal operating range, was used to estimate the drug-substance quality distribution and process capability (Figure 8.1, Table 8.1). Capability is evaluated one-sided for impurities (upper limit) and for viral clearance (lower requirement), and two-sided for target-range attributes.

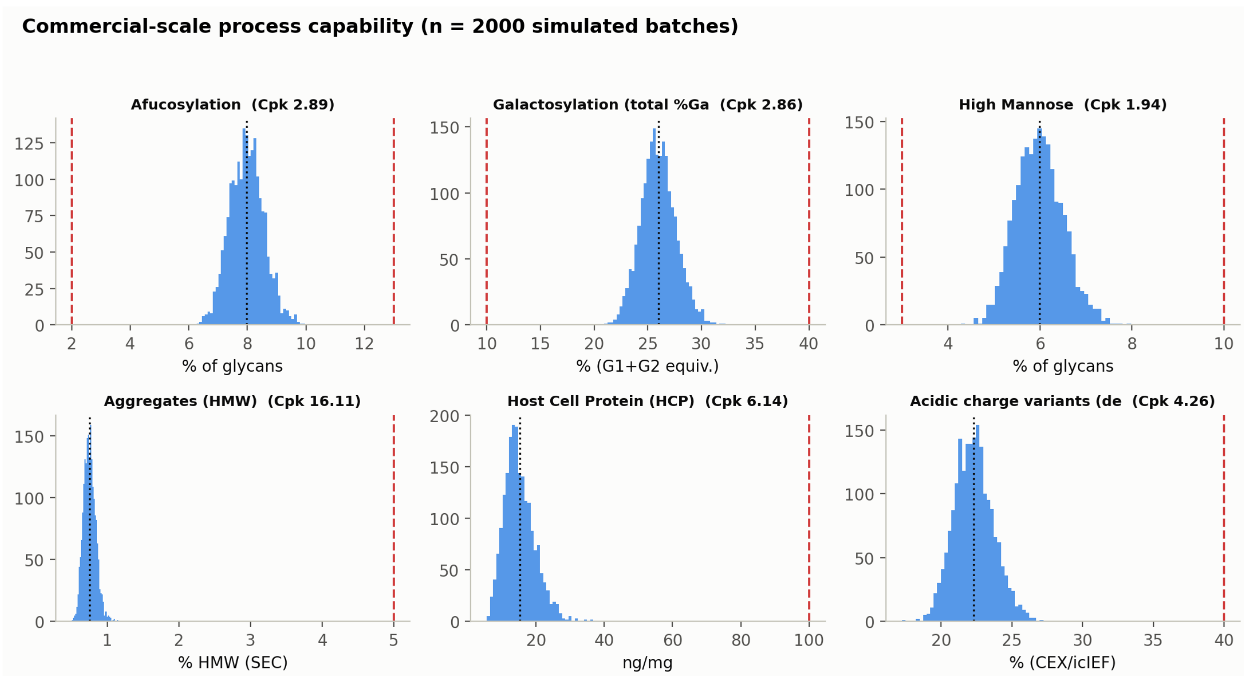

Figure 8.1: Commercial-scale drug-substance CQA distributions vs acceptance limits, annotated with process capability (Cpk).

Table 8.1: Commercial-scale process capability by CQA (n = simulated batches).

CQA	Crit.	Spec	Acceptance	Mean ± SD	Cpk
Afucosylation	H	two_sided	2–13	7.98 ± 0.58	2.89
Galactosylation (total %Gal)	H	two_sided	10–40	26 ± 1.6	2.86
High Mannose	VH	two_sided	3–10	5.99 ± 0.51	1.94
Aggregates (HMW)	H	upper	0–5	0.753 ± 0.088	16.11
Acidic charge variants (deamidation)	VL	upper	18–40	22.3 ± 1.4	4.26
Host Cell Protein (HCP)	M-H	upper	0–100	15.2 ± 4.6	6.14
Residual DNA	VL	upper	0–0.001	0.0001 ± 0	10.81

CQA	Crit.	Spec	Acceptance	Mean ± SD	Cpk
Leached Protein A	L-M	upper	0–5	0.003 ± 0.0006	2785.30
Viral clearance — XMuLV (enveloped)	VH	lower	16.7–30	19.5 ± 0.61	1.53
Viral clearance — MVM (parvovirus)	VH	lower	8.6–20	10.4 ± 0.4	1.51

The minimum capability across all CQAs is **Cpk = 1.51** (all CQAs 1.33). A representative PPQ campaign of 5 batches (Table 8.2) meets all acceptance criteria.

Table 8.2: Representative PPQ-campaign drug-substance results.

Batch	Yield	aFuc %	Gal %	HMW %	HCP (ng/mg)	XMuLV LRV	MVM LRV
PPQ-001	0.782	7.44	26.6	0.817	17.2	19.7	9.9
PPQ-002	0.754	8.51	28	0.614	14.5	19.5	9.87
PPQ-003	0.732	7.42	24.3	0.596	15.9	19.7	10.6
PPQ-004	0.763	7.94	25.1	0.927	14.2	20.1	10.4
PPQ-005	0.838	6.8	25.8	0.664	7.96	19.9	9.93

Viral clearance is delivered by three orthogonal steps (Figure 8.2, Table 8.3): cumulative clearance of **18.87 log (XMuLV)** and **10.03 log (MVM)** exceeds the requirements of > 16.7 and > 8.6 with margin. Product step-yield across the train is **83.2%** (Figure 8.3).

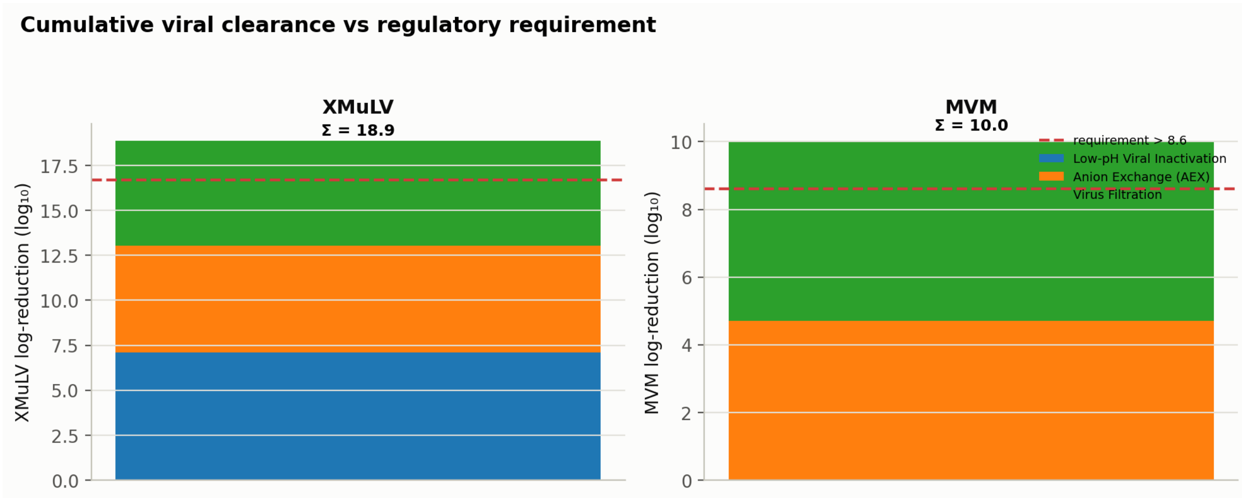

Figure 8.2: Modular viral clearance by step for XMuLV and MVM; cumulative clearance exceeds the total requirement.

Table 8.3: Viral log-reduction by step and cumulative total.

Step	XMuLV	MVM
Low-pH Viral Inactivation	7.10	0.00
Anion Exchange (AEX)	5.95	4.71
Virus Filtration	5.82	5.32
Cumulative	18.87	10.03

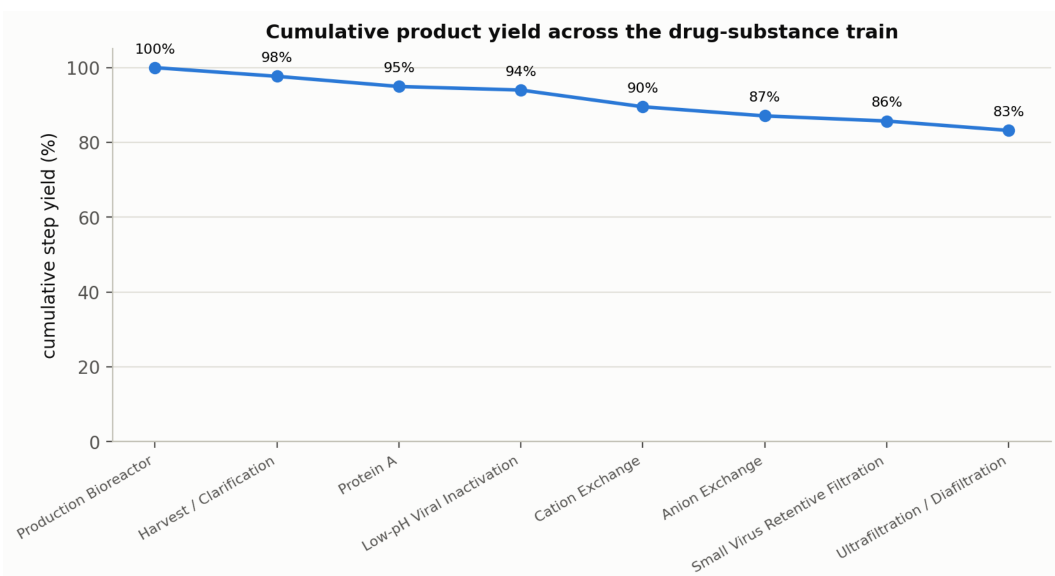

Figure 8.3: Cumulative product step-yield across the drug-substance train.

9 Control strategy

The control strategy is the sum of the individual CQA control strategies (ICH Q10), built from the characterization understanding above. Each CQA is controlled by a defined combination of input-material controls, procedural controls, process-parameter set-points and proven acceptable ranges, in-process controls and release testing (Table 9.1). Glycan CQAs are controlled predominantly at the bioreactor; aggregate at CEX (with the low-pH hold bounded); HCP across Protein A, CEX and AEX; and viral safety by three orthogonal clearance steps plus the low-pH inactivation.

Table 9.1: Overall control strategy — principal controls per CQA.

CQA	Principal process controls	Testing
Afucosylation / high mannose / galactosylation	Bioreactor pH, temperature, CO , osmolality, duration (design space)	Released glycan map; in-process culture control
Aggregates (HMW)	CEX (load, conductivity, elution pH, stop-collect); low-pH hold bounded	SEC at CEX pool and release
Acidic charge variants	Bioreactor set-points; CEX/AEX collection	Charge-variant assay at release
Host cell protein (HCP)	Protein A (load, elution pH); CEX; AEX (load pH, conductivity)	HCP ELISA at AEX pool and release
Residual DNA	Protein A + AEX clearance (modular)	qPCR at release
Leached Protein A	CEX + AEX clearance	Leached Protein A assay at release
Viral safety (XMuLV / MVM)	Low-pH inactivation; AEX; virus filtration (validated ranges)	Validated clearance; in-process bioburden/endotoxin

10 Conclusions and readiness for Stage 2

The A-Mab drug-substance process is well understood and robust:

- Every CQA meets its acceptance criterion at set-point, with a minimum capability of **Cpk = 1.51** across 2,000 simulated commercial batches.
- The relationships between process parameters and CQAs are quantified by DoE, design spaces are established for the bioreactor and the chromatography and viral-safety steps, and every parameter range is justified by data.
- 37 parameters are classified; 21 are critical (CPP/WC-CPP) with a defined control strategy, and the post-characterization FMEA shows the median risk priority number falling by ~85% relative to the initial assessment.
- Viral clearance exceeds requirements with margin (18.87 / 10.03 log for XMuLV/MVM).

The process design (Stage 1) is complete. The characterized ranges, design spaces and control strategy provide the basis for the Stage 2 Process Performance Qualification protocol, acceptance criteria and sampling plans. The process is recommended for progression to PPQ.

Reproducibility

Every figure, table and number in this report is generated from a seeded process model (master seed 20240724). The report is regenerated end-to-end with `make all` (see the repository README): the model produces the datasets (`outputs/data/`) and figures (`outputs/figures/`), and Quarto renders this document to Word and PDF from the same sources.

References

CMC Biotech Working Group. 2009. "A-Mab: A Case Study in Bioprocess Development." CASSS.

International Council for Harmonisation. 2009. "ICH Q8(R2): Pharmaceutical Development." ICH.

———. 2012. "ICH Q11: Development and Manufacture of Drug Substances." ICH.

———. 2023. "ICH Q9(R1): Quality Risk Management." ICH.

International Society for Pharmaceutical Engineering. 2023. "Good Practice Guide: Practical Implementation of the Lifecycle Approach to Process Validation." ISPE.

Parenteral Drug Association. 2013. "Technical Report No. 60: Process Validation — a Lifecycle Approach." PDA.